

Shanrong Wu

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, Illinois

Bachelor of Science in Computer Science — GPA: 3.88

Expected Graduation Date: May 2028

Relevant Coursework: Introduction to Computer Science I, Introduction to Computer Science II (Honors), Discrete Structures, Data Structures, Computer Architecture, Software Design Lab

PROJECTS

HackIllinois 2026 – HackAstra | *React, Python, Flask, Gemini API, Supermemory*

Feb 2026

- Led a team of 3 to develop an AI inspection software application for Caterpillar Track, automating daily inspections through audio, video, and image-based input.
- Integrated the Gemini API to deliver real-time AI-generated inspection feedback during the inspection workflow.
- Built full stack features using React and Flask, including inspection processing and reporting.
- Implemented a machine inspection history to store and display previous inspection results for machines.
- Improved report generation by debugging inspection output and creating functionality to consolidate multiple same day inspections into one unified report in daily checklist and history.

Raytracer in Rust | *Rust*

Oct 2025 – Jan 2026

- Developed a ray tracing application in Rust with a 3 person team, utilizing Rust crates for rendering and systems level performance improvements.
- Designed and implemented geometry processing utilities to convert 3D coordinates and STL mesh data into triangle objects for ray object intersection testing using the Möller–Trumbore algorithm.
- Optimized runtime by about 10x with parallel processing, multithreading, and denoising techniques.
- Created scenes with STL based models and geometric objects to check rendering accuracy and scene composition.

FRC Path Following Robot | *Java, WPILib Library*

Jan 2023 – Feb 2023

- Developed an autonomous path following system for an FRC robot using Java and the WPILib framework, increasing the number of autonomous paths available for competition.
- Tuned PIDF control loops and motor configuration to reduce tracking error and improve autonomous accuracy.
- Improved the process for creating, testing, and deploying autonomous paths, making development more efficient.

EXTRACURRICULARS

SIGrobotics - F1tenth

Sep 2025 – Present

Simulation/Programmer

Urbana, Illinois

- Develop autonomous driving functionality for the F1TENTH racecar in the F1TENTH Gym simulation using C++, ROS 2, PID control, and path planning/control algorithms across multiple track layouts.
- Support development for the physical racecar by working with 2D LiDAR, stereo vision, and Jetson Nano.
- Participate in meetings to collaborate on project progress and continued learning of autonomous racing concepts.
- Help define project goals and contribute to software development for both simulated and physical racecar testing.
- Engineered a local planner in Python and ROS2 Humble that integrated global planner output with robot pose data to generate waypoints for pure pursuit based navigation.

FIRST Robotics Competition

Aug. 2021 – May 2025

Software Lead, Technician, Programmer, Electrical

Bradley, Illinois

- Mentored team members in Java and LabVIEW through hands on demonstrations and programming activities.
- Resolved electrical and software issues under competition time restraint pressure to maintain robot reliability, reducing match time failures and improving overall performance and placement each year.
- Upgraded software and hardware systems from deprecated software to updated platforms and current standards.
- Drove the migration from LabVIEW to Java by demonstrating advantages in scalability and long term support.
- Researched swerve drive kinematics, geometric control, and vision-based pose estimation to improve robot localization and enable autonomous execution of diverse tasks.

TECHNICAL SKILLS

Languages: C++, Java, Python, Rust, LabVIEW, MIPS Assembly, JavaScript, TypeScript, HTML/CSS

Frameworks: React, Next.js, TailwindCSS, ROS2

Developer Tools: Linux, Git, GitHub, GitLab, Docker, VS Code, IntelliJ, Supabase